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Multiscale and Multilayer Contrastive Learning for Domain Generalization ($M^2 - CL$)

Người thực hiện:

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Presentation Agenda

1. Introduction
2. The $M^2 - CL$ Architecture
3. Experimental
4. Qualitative Visualizations
5. Conclusion
6. Appendix

Introduction to Domain Generalization (DG)

The Core Objective

Train a parametric model $f(\cdot;\theta)$ on a set of N available **source domains** $\mathcal{S} = \{S_1, S_2, \ldots, S_N\}$ such that it generalizes robustly to K completely **unseen target domains** $\mathcal{T} = \{T_1, T_2, \ldots, T_K\}$.

The Strict Constraint vs. Domain Adaptation

- **Domain Adaptation (DA):** Has access to unlabeled target domain data during training.
- **Domain Generalization (DG):** Zero access to target-domain data or target labels during the optimization cycle.

The ERM Trap

- Models minimize error by taking the **path of least resistance**.
- Exploits superficial shortcuts (textures, backgrounds).

Domain Shift Collapse

- **Photo → Sketch:** Context and style cues completely vanish.
- Result: Catastrophic accuracy drop.

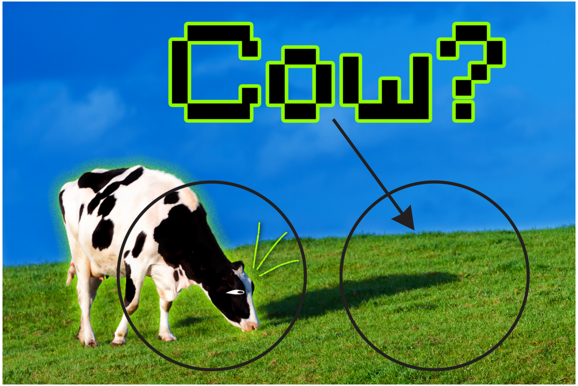


Fig: ERM relies on "Grass" (Shortcut) instead of the actual "Cow" geometry.

Information Bottleneck in Deep ConvNets

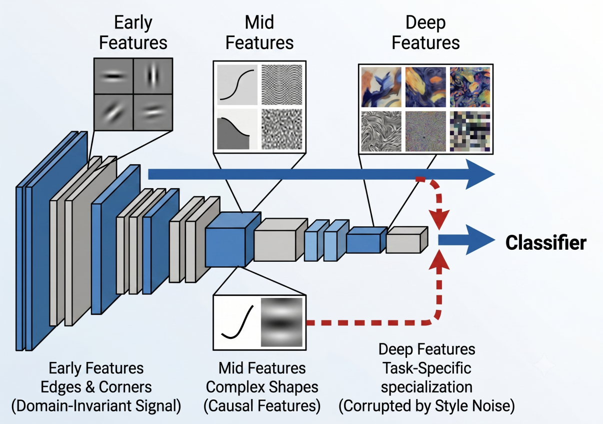
The Last-Layer Problem

- Standard CNNs use **deepest features** only.

The Invariant Signal

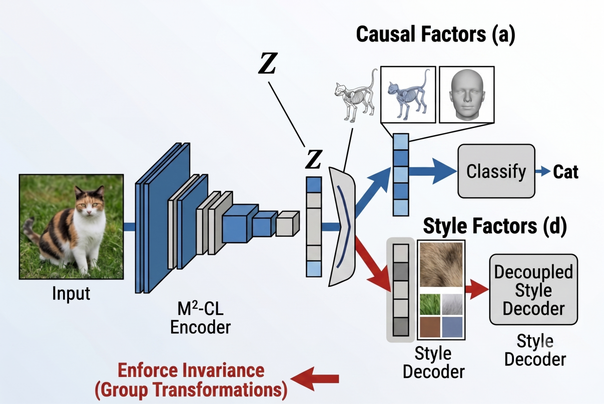
- **Early/Mid Layers:** Hold shapes and edges.

$\Longrightarrow$ Corrupted by style and filters out raw structure details.



Latent Space Split: $X \approx (a, d)$

- **Causal** (a)**:** Class-specific **geometry**(Invariant).
- **Style** (d)**:** Domain-specific **context** and texture (Noise).

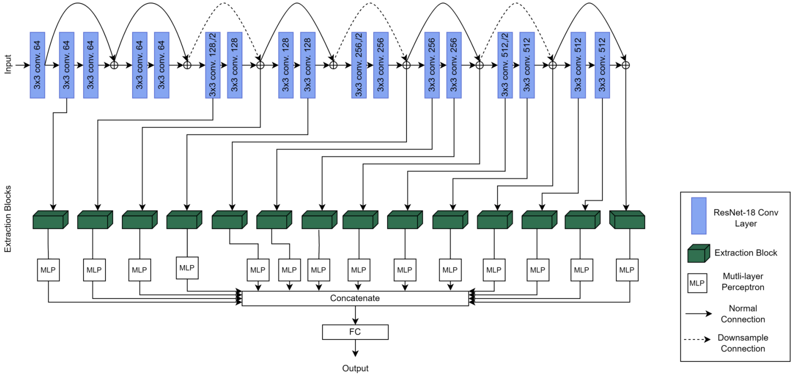


The DG Goal

Isolate **Causal** a from **Style** d to ensure the model only relies on invariant geometry for prediction.

Framework Concept

Leverages parallel multi-scale concentration blocks mapped across multiple intermediate layers of a standard backbone network.



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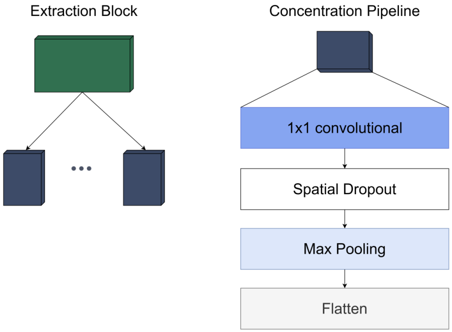
Extraction Block



Concentration pipeline layout.

Internal Sequence Mechanics

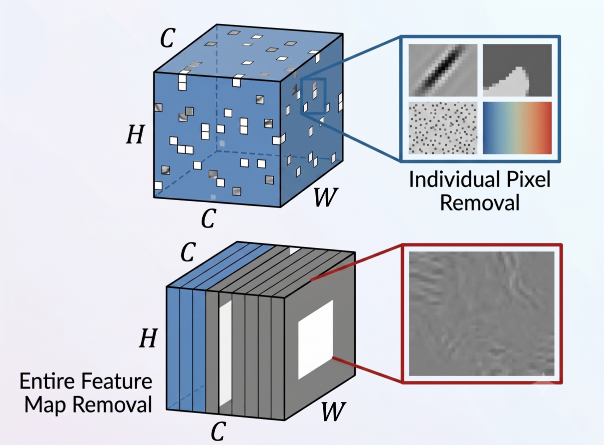
1. 1×1 **Convolutional Layer:** Compresses high-dimensional channels c down to $\lfloor c/r \rfloor$ using a reduction factor r.
2. **Spatial Dropout Layer:** Mitigates pixel correlation leakage.
3. **Multi-Scale Max Pooling:** Captures spatial features before reduction.
4. **MLP Projection Vector:** Maps pooled features into vectors via MLPs.

Vanilla Dropout (CNN Failure)

- Drops **individual pixels** randomly.
- Result: Info leaks; regularization fails.

Spatial Dropout (Our Solution)

- Drops **entire feature maps** $(H \times W)$.
- Forces network to find alternative cues.
- **Result:** Blocks local style shortcuts.



Multi-Scale Concentration & Hypercolumn Construction

Spatial Scaling Allocations

- **Early Hooks:** Pooling sizes optimized for 8×8, 4×4, and 2×2.
- **Late Hooks:** Deeper fields adapt to 7×7 and 3×3 dimensions.

Concatenation Protocol

Flattened multi-scale vectors are concatenated into a **hypercolumn vector** for final FC layer classification.

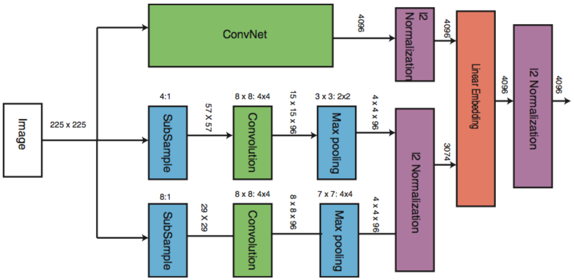


Fig: Multi-scale pooling pipelines merging into a unified Hypercolumn vector.

The M2-CL Loss Framework

Geometric Constraints

- **Pull:** Same class, cross domains.
- **Push:** Different classes, same style.

Matrix Parallelization

- Prevents memory bottlenecks.
- Stacks batch: $U^{(l)T}U^{(l)}$.
- Denominator computed **once**.

Alignment Measure ($\|u\| = 1$)

$$p^{(l)}(c) = \frac{\sum_{i,j \in N_c} \exp(u_i^T u_j / \tau)}{\sum_{i,a \in N_b} \exp(u_i^T u_a / \tau)}$$

Total Objective

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \sum_{l=1}^{M} \mathcal{L}^{(l)}$$

$$\mathcal{L}^{(l)} = -\sum_{c=1}^{C} \log p^{(l)}(c)$$

Technical Specifications & Hyperparameters

Optimizer Configurations

- Base Optimization: Stochastic Gradient Descent (SGD) for 30 epochs.
- Learning Rate: 0.001, Batch Size Allocation: 128.
- Default Extraction Blocks Settings: Reduction factor $r = 4$.
- Default Loss Balancing: Weight $\alpha = 0.01$, Temperature $\tau = 1.0$.

VLCS Dataset

- **Size:** 10,729 images | **Classes:** 5
- **Domains (4):** PASCAL, LabelMe, Caltech, SUN.

PACS Dataset

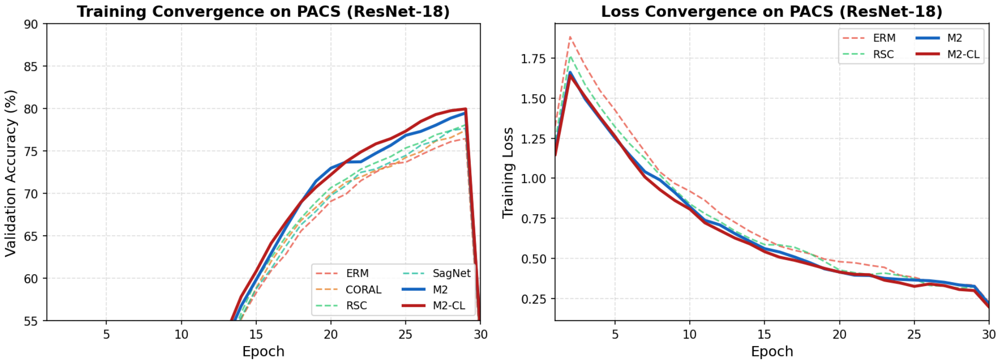
- **Size:** 9,991 images | **Classes:** 7
- **Domains (4):** Photo, Art Painting, Cartoon, Sketch.

Office-Home Dataset

- **Size:** 15,588 images | **Classes:** 65
- **Domains (4):** Art, Clipart, Product, Real-World.

(a) VLCS (b) PACS (c) Office-Home

Training Convergence on PACS (ResNet-18)



M^2-CL converges faster and plateaus **higher** than all baselines. The contrastive regularizer suppresses loss spikes caused by domain-conflicting gradients.

Dataset Table : PACS and VLCS (ResNet-18)

Top-1 accuracy on PACS and VLCS, ResNet-18.

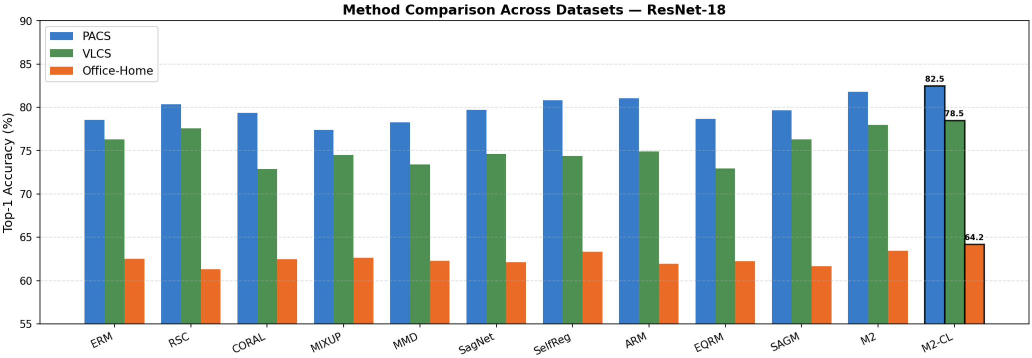
Method	Art	Cartoon	Photo	Sketch	Avg	Caltech	Labelme	Sun	Voc	Avg
ERM	76.53	74.80	94.95	67.89	78.54	96.87	65.03	71.62	71.60	76.28
RSC	79.26	76.12	95.07	70.88	80.34	97.68	66.33	73.53	72.75	77.57
CORAL	78.55	72.74	92.24	73.93	79.36	95.16	60.73	70.66	65.00	72.89
MIXUP	79.79	72.81	93.11	63.97	77.42	93.44	60.25	**74.36**	69.87	74.48
MMD	77.75	73.35	92.34	69.74	78.29	97.28	60.62	70.48	65.21	73.40
SagNet	78.71	74.90	94.79	70.48	79.72	95.76	61.22	69.61	71.90	74.62
SelfReg	79.46	76.32	**95.93**	71.60	80.83	92.53	62.56	70.27	72.15	74.38
ARM	79.77	75.33	94.43	<u>74.76</u>	81.07	95.86	60.19	70.66	72.83	74.89
EQRM	78.61	72.38	92.48	71.13	78.65	96.17	60.57	70.44	64.66	72.96
SAGM	78.22	75.14	94.79	70.46	79.65	96.37	66.22	71.14	71.35	76.27
M2	<u>80.64</u>	<u>76.91</u>	95.50	74.20	<u>81.81</u>	<u>98.24</u>	<u>67.02</u>	73.80	<u>72.92</u>	<u>78.00</u>
M2-CL	**81.04**	**77.48**	<u>95.70</u>	**75.81**	**82.51**	**98.69**	**67.48**	<u>74.05</u>	**73.76**	**78.50**

Dataset Table : Office-Home (ResNet-18)

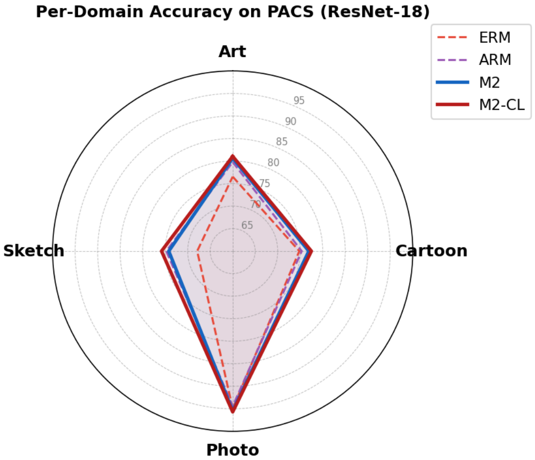
Top-1 accuracy on Office-Home, ResNet-18.

Method	Art	Clipart	Product	Real-World	Avg
ERM	56.12	47.88	<u>72.47</u>	73.58	62.51
RSC	56.00	46.09	70.85	72.32	61.31
CORAL	55.46	49.39	71.75	73.28	62.47
MIXUP	56.37	<u>49.81</u>	70.87	73.61	62.66
MMD	55.67	48.52	71.68	73.19	62.27
SagNet	55.75	47.74	71.71	73.17	62.09
SelfReg	57.56	49.69	72.43	73.72	63.35
ARM	54.47	48.45	71.14	<u>73.79</u>	61.96
EQRM	56.08	48.34	71.57	72.94	62.23
SAGM	55.79	46.51	71.16	73.15	61.65
M2	<u>58.04</u>	49.30	**73.04**	73.50	<u>63.47</u>
M2-CL	**58.77**	**50.86**	72.20	**74.91**	**64.18**

Method Comparison — All Datasets (ResNet-18)



Per-Domain Analysis on PACS (ResNet-18)



Key Observations

- M^2-CL achieves the best overall balance across all domains.
- Largest improvement: **Sketch** (+7.9 pp vs. ERM), indicating robustness to severe style shifts.
- Gains on **Art** suggest better cross-domain feature alignment.

Quantitative Results: Main Takeaways

Average Accuracy

- **ResNet-18:** M^2-CL reaches **82.51** on PACS, **78.50** on VLCS, and **64.18** on Office-Home.
- The average score improves over the strongest baseline in each benchmark summary.

Interpretation

- M^2 improves feature representations through multi-scale reuse.
- M^2-CL aligns same-class features across domains.
- Ablation studies identify the sources of these gains.

Table: Ablation Study on PACS and VLCS

Ablation study on PACS and VLCS, ResNet-18.

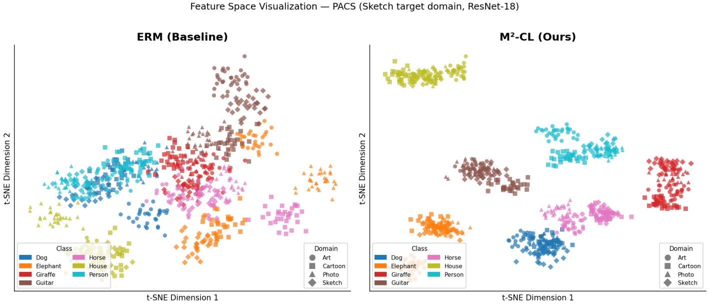
pipe	r	drop	loss	A	C	P	S	PACS Avg	C	L	S	V	VLCS Avg
c	2	-	-	72.34	72.29	94.19	69.51	78.23	93.87	60.91	65.89	61.91	71.22
c	4	-	-	74.38	72.74	90.13	69.71	77.21	94.14	59.01	67.82	65.42	69.62
c	6	-	-	75.72	70.12	94.26	73.33	77.92	91.93	56.81	67.28	64.10	68.95
c	2	yes	-	75.67	73.29	92.16	75.95	77.93	93.73	57.72	66.34	62.50	71.74
c	4	yes	-	74.36	74.05	92.57	76.45	79.37	96.07	56.28	67.07	64.30	65.81
c	6	yes	-	75.97	72.77	93.01	74.73	76.81	92.94	59.73	66.15	64.51	70.09
p	2	-	-	76.09	55.88	68.35	60.92	64.40	95.60	56.97	66.25	73.15	73.96
p	4	-	-	74.72	74.62	92.01	70.32	77.64	95.20	58.10	66.20	64.90	70.29
p	6	-	-	75.38	69.99	91.21	69.47	77.77	95.23	56.81	69.13	65.34	70.55
p	2	yes	-	74.61	72.39	87.08	73.95	78.88	95.73	58.33	68.44	63.74	70.49
p	6	yes	-	76.73	73.98	93.41	73.56	79.84	93.19	56.05	66.62	64.24	70.43
p	4	yes	-	76.39	75.98	95.58	74.49	79.15	94.54	59.77	67.26	74.55	73.14
p	4	yes	yes	80.43	75.43	96.87	74.42	80.10	96.09	61.15	69.05	75.70	74.15

Core Ablation Analysis

Table: Reference Insights (PACS / VLCS, ResNet-18)

- **Best full configuration:** $p, r = 4$, $drop = yes$, $loss = yes$ reaches **80.10** PACS Avg and **74.15** VLCS Avg.
- **Contrastive loss contribution:** adding the loss to the same $p, r = 4$, $drop = yes$ setting improves PACS from **79.15** to **80.10** and VLCS from **73.14** to **74.15**.
- **Spatial dropout effect:** for $p, r = 4$, dropout improves PACS from **77.64** to **79.15** and VLCS from **70.29** to **73.14**.

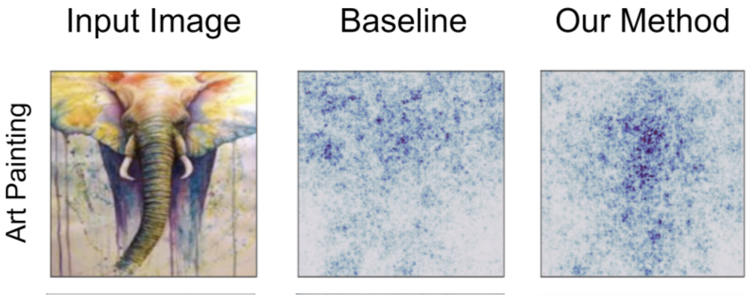
Feature Space Visualization: ERM vs M^2-CL



Left (ERM): Same-class features from different domains are dispersed — the model learns domain-specific patterns. **Right (M^2-CL):** Contrastive regularization collapses domain scatter within each class while separating inter-class clusters.

Methodology

Visualizes the gradients of the class score function relative to input image pixels. Darker pixel weights highlight the regions that contribute most to final classification decisions.



Saliency maps comparison on the PACS dataset.

Summary of Key Contributions

Core Achievements

- **Architecture:** Successfully combines multi-scale, multi-layer intermediate features without propagating domain-specific styles through the network.
- **Loss Optimization:** Introduces a layer-wise contrastive loss that enforces domain invariance across all extraction levels.
- **Empirical Validation:** Outperforms previous domain generalization methods and sets a new state-of-the-art across four major benchmark datasets.

Limitations & Future Directions

Current Computational Drawbacks

- **Memory Overhead:** Concatenating multiple intermediate feature maps creates dense hypercolumns before the classification head, increasing memory usage.
- **Batch Size Dependency:** Like most contrastive methods, M^2-CL requires large training batches to construct effective positive/negative pairs.

Future Research Avenues

Explore cross-layer attention mechanics, build causal inference systems, and evaluate alternative similarity metrics like KL divergence to refine latent distribution shapes.

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Dataset Table: PACS and VLCS (ResNet-50)

Top-1 accuracy on PACS and VLCS, ResNet-50.

Method	Art	Cartoon	Photo	Sketch	Avg	Caltech	Labelme	Sun	Voc	Avg
ERM	83.54	78.47	96.51	73.64	83.04	97.01	62.22	71.79	76.09	76.78
RSC	82.08	**80.26**	96.51	75.45	83.57	**97.81**	64.44	74.60	75.53	78.10
CORAL	83.94	79.27	94.37	**81.98**	84.89	97.01	61.34	75.26	71.74	76.34
MIXUP	85.16	77.40	96.23	78.26	84.26	96.94	62.33	73.40	72.32	76.25
MMD	82.21	74.70	92.89	69.00	79.70	97.48	63.85	75.49	71.65	77.12
SagNet	85.23	78.47	93.20	77.71	83.65	94.14	60.43	70.49	69.96	73.75
SelfReg	83.33	79.06	93.16	77.24	83.20	92.49	60.73	66.80	73.58	73.40
ARM	82.14	79.10	92.97	73.02	81.81	96.01	63.98	66.26	71.79	74.51
EQRM	84.98	77.74	91.26	74.88	82.21	94.35	60.80	66.35	72.94	73.61
SAGM	81.35	78.99	93.24	79.39	83.24	97.39	64.23	69.30	76.17	76.77
M2	<u>86.02</u>	79.90	<u>97.03</u>	81.50	<u>86.11</u>	97.52	<u>64.77</u>	<u>76.02</u>	<u>76.72</u>	<u>78.76</u>
M2-CL	**86.48**	<u>80.05</u>	**97.99**	<u>81.70</u>	**86.56**	<u>97.60</u>	**64.86**	**76.65**	**77.26**	**79.09**

Dataset Table: Office-Home (ResNet-50)

Top-1 accuracy on Office-Home, ResNet-50.

Method	Art	Clipart	Product	Real-World	Avg
ERM	63.43	51.32	76.45	<u>78.13</u>	67.33
RSC	62.64	51.55	75.11	77.34	66.66
CORAL	61.27	<u>54.34</u>	73.35	77.83	66.70
MIXUP	63.23	51.91	<u>77.10</u>	76.39	67.16
MMD	59.63	49.37	75.28	74.71	64.75
SagNet	61.41	51.85	72.85	73.79	64.97
SelfReg	59.95	51.22	71.28	74.52	64.24
ARM	54.23	52.14	71.73	71.26	62.34
EQRM	63.52	52.32	73.28	76.45	66.39
SAGM	61.01	53.27	74.81	77.00	66.52
M2	<u>65.35</u>	53.90	76.80	**78.74**	<u>68.70</u>
M2-CL	**66.46**	**54.82**	**78.24**	77.85	**69.34**

Table: Sensitivity to Temperature τ

Sensitivity analysis for τ in M2-CL, ResNet-18.

τ	PACS	VLCS
0.01	78.10	76.15
0.1	80.03	77.27
0.2	81.37	74.71
0.4	82.07	72.03
0.6	83.70	74.15
0.8	84.67	75.21
1.0	80.07	76.24
1.2	81.34	75.11
1.4	83.14	75.69
1.6	82.17	76.84
1.8	80.51	74.66
2.0	78.72	74.99
10.0	79.92	74.04
100.0	81.84	73.74

Table: Sensitivity to Loss Weight α

Sensitivity analysis for α in M2-CL, ResNet-18.

α	PACS	VLCS
0.00	82.76	72.36
10e-5	80.41	73.56
10e-4	81.17	73.71
10e-3	82.29	74.38
10e-2	79.70	73.39
10e-1	64.04	55.96

Sensitivity Analysis: Loss Scaling (α & τ)

Tables 6–7 Insights: Hyperparameter Tuning

- **Temperature optimum:** PACS peaks at **84.67** with $\tau = 0.8$, while VLCS peaks at **77.27** with $\tau = 0.1$.
- **Default-scale behavior:** $\tau = 1.0$ remains competitive with **80.07** PACS and **76.24** VLCS.
- **Loss-weight sensitivity:** $\alpha = 10e^{-3}$ gives the best VLCS score, **74.38**, while $\alpha = 0$ gives the highest PACS score, **82.76**.

Alpha Bounds Limits

$$\alpha = 10e^{-1} \implies \text{PACS } 64.04, \quad \text{VLCS } 55.96$$

A large contrastive weight dominates cross-entropy optimization and causes a sharp drop compared with the best scores in Table 7.